

Comparative Study of Machine Learning Algorithms for Breast Cancer Prediction

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Abstract—Breast cancer is the most common cancer occurring in women and is estimated to be 270,000 new cases diagnosed in 2019. This is the reason detection software is needed to detect it before it gets fatal. By the use of machine learning algorithms, software can be made to detect this dangerous cancer and treat it before it can cause fatality of the patient. It is also the most frequently occurring cancer among Indian women and the chance for survival of a woman suffering from breast cancer is 50%. For Breast Cancer Detection many machine learning algorithms can be used. In this paper, 2 machine learning algorithms is proposed to compare namely **Logistic Regression** and **Decision Tree algorithm** on the **Wisconsin (Diagnostic) Data Set** and use the algorithm with the best accuracy for predicting Breast Cancer.

Keywords—Breast Cancer, Machine Learning Algorithm, Logistic Regression, Decision Tree, Wisconsin (Diagnostic) Data Set.

I. INTRODUCTION

Breast Cancer is one of the main reasons for the demise of women. It is the second dangerous cancer after lung cancer. Like any other Cancer, Breast Cancer begins when healthy cells change and starts to grow in a disordered manner, forming a mass of cells called a tumor. A tumor can be benign or cancerous. A cancerous tumor is called malignant and they can grow and spread to other parts of the body of the patient. A benign tumor is a tumor that can grow in a particular part of the body but it does not spread to the other parts of the body. Many challenges are faced by a woman fighting against breast cancer which includes pain during radiation therapy and chemotherapy, the huge cost in terms of money that comes with it and, much more so it becomes very essential to predict breast cancer as soon as possible. There are pieces of evidence (as suggested by WHO) that state that women who consume drinks that contain alcohol, have above average birth-weight and above-average height attained when they are an adult are more at risk to develop breast cancer. It is also suggested that physically active women, eat whole grains, vegetables, fruits, and consume less red meat, alcoholic drinks and, sugar-sweetened drinks are at lower risk of developing breast cancer.

The following graphs explain the statistics related to breast cancer such as most vulnerable age of developing breast cancer, average cases per year to the age and much more [3]:

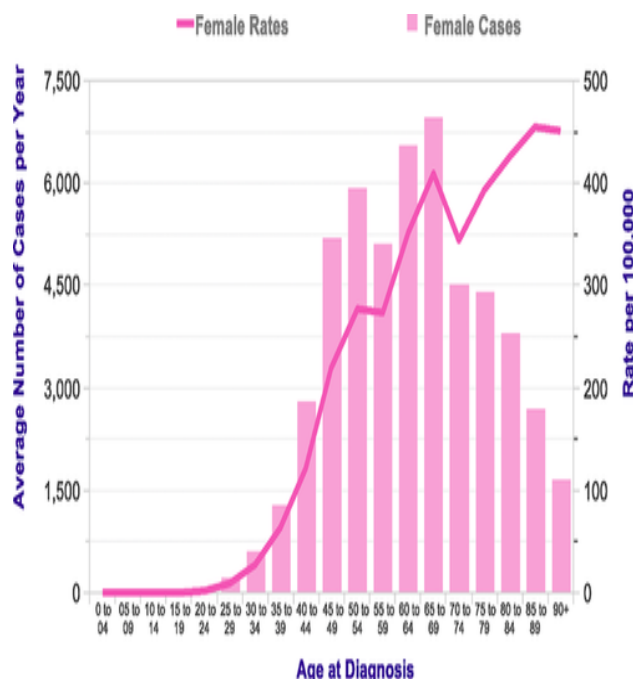


Fig. 1. Statistics Related to Breast Cancer

The following are some images that represent the difference between a normal breast and a cancerous breast [4].

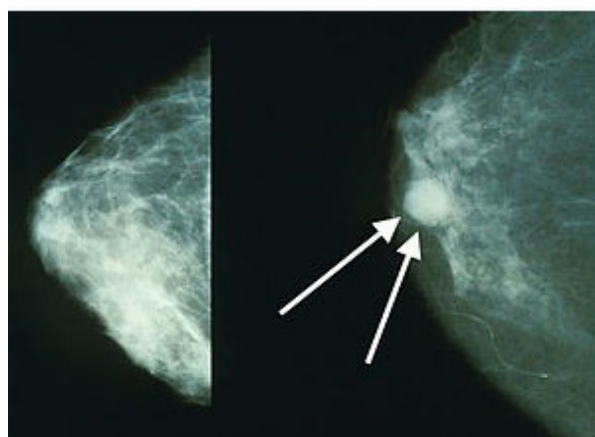


Fig. 2. Normal breast (left) and cancerous breast (right)

By the use of machine learning algorithms and techniques, if the cancer is benign or malignant will be predicted by looking at the symptoms and attributes of a person suffering from any one of them and comparing them to the symptoms and attributes of the potential victim. Malignant cancer is a more dangerous type of breast cancer and detecting it at an early phase will result in better treatment hence less harm to the patient. Also, early treatment is less expensive. Machine Learning, with its ability to extract main features from complex datasets, is largely used to predict breast cancer. Application of these machine learning techniques in the medical field is of great importance as a disease can be predicted in an initial stage which can help us reduce the cost of medication that goes with it, help aid people's health, predict mostly accurate outcomes and help upgrade the healthcare value and save people's lives.

II. Related Work

Lots of breast cancer research has been reported in the past, and most of them turned up with good classification accuracies. One such paper is named tumor size classification of breast thermal images using fuzzy C-Means algorithm [5]. Here they used the concept of grouping or clustering by the use of C-Means algorithm. The method uses the data grouping technique by classifying based on color component. They proposed a method of box counting of fractal algorithms to extract the features of the image and then cluster the data obtained from those breast cancer thermography images to determine the stage of cancer, using the Fuzzy C-Means clustering algorithm. To do the same they first preprocessed the data by converting RGB images to Grayscale and then those to Binary form. The RGB image stands for "Red Green Blue" image. It refers to three shades of light that when mixed together can create different colors. Combining red, green and blue shades is the standard method for producing colored images. Grayscale images are those which are black and white. Last but not the least converting the image into Binary form means pixels with a grey level above a certain threshold level are set to 1 (i.e. White), whilst the rest are set to 0. So this process removes all the color information and leaves only luminance of each pixel. Then the features were extracted by the use of box counting. In the Box counting method, data is taken and analyze it by breaking it (dataset, image, object, etc) into small pieces. Which are box shaped and then analyze these smaller pieces on a smaller scale. It also has applications in related fields such as multifractal and lacunarity analysis. And finally, the classification was done through Fuzzy C-Means. The advantage of doing this was they were only using thermal pictures with some data to predict the cancer but the disadvantage lied in the accuracy of the result which was not up to the mark.

Another research paper called Breast Cancer Diagnosis using Deep Learning Algorithm [6] used deep learning algorithms to diagnose breast cancer. They used the convolutional neural network for the diagnosis of the cancer. Deep learning, which is a subgroup of machine learning where neural networks algorithms like the human brain are used to learn from a large amount

of data which helps us solve complex problems. To preprocess the data they used techniques such as Label Encoder Method which is used to convert the labels which are non-numeric into the numeric form so it can be used in machine learning models. So that machine learning algorithms can assign the labels in a better way. This step is quite an important preprocessing step for performing supervised learning. Normalization and Standard Scalar Method, which assumes your data is normally distributed within each feature and will scale them such that the distribution is now centered on 0, with a standard deviation of 1. If data is not normally distributed, this is not the best scalar to use as it can affect the results. Then they used a deep learning neural network algorithm which contains a series of algorithms that tries to recognize some useful relationships in the dataset that copies the working of the human brain to diagnose cancer.

Another research called Breast Cancer Diagnosis Using Adaptive Voting Ensemble Machine Learning Algorithm [7]. They use the concept of ensemble method for diagnosing breast cancer using neural networks and logistic algorithms. Ensemble methods use multiple models to produce improved results. They usually produce more accurate results than a single model would. The easiest ensemble models are 'Voting model' and 'Averaging model' as they both are easy to understand and implement. Averaging model is used for regression and the Voting model is used for classification. To preprocess the data they used the Standardization method and selected the features according to 'Univariate feature selection' that selects the best features on univariate statistical tests. It compares each feature to the target variable to find if the given feature can be used to predict our target variable or not. This is known as analysis of variance (ANOVA). That is why it is called 'univariate'. Each feature has its test score. And finally, used the Neural Network model to diagnose the cancer. The advantages of the above 2 techniques using neural network were that the accuracies of prediction were improved but the disadvantage was the results consumed greater time to be generated as neural network is a complex algorithm.

Another paper called the Breast Cancer Diagnosis using an Unsupervised Feature Extraction Algorithm Based on deep learning [8]. The main objective of this research was to predict breast cancer using unsupervised deep learning based on feature extraction strategy. The goal of unsupervised learning is to create models that can be trained with the use of little data. Today Deep Learning models are trained on large supervised datasets which means that, for each data, there is a corresponding label. Then Feature extraction is used which is a type of dimensionality reduction method that shows the important areas of the image as a compact feature vector. This is used when the size of the images is large and require a reduced feature representation. This is done so that image retrieval, image matching, etc can be done quickly. After extracting the required features they used the SAE-SVM model (support vector machine) to diagnose the cancer. Now the Support Vector Machine (SVM) comes under supervised machine learning that is used for regression and classification challenges. It's mostly used in classification problems. Support Vectors are nothing but

the coordinates of an individual observation. The advantage was the ease of extracting features but the problem was not up to the mark accuracies and time consumption for result.

Another paper known as Artificial Intelligent Models for Breast Cancer Early Detection [9]. The main objective of this research was to use Artificial Intelligence (AI) to help them in processing data for getting faster and accurate results and information. Artificial intelligence (AI) is nothing but simulating the human brain or intelligence in machines so that they think like humans to make decisions and perform actions. The term can also be associated with any machine that has traits like a human mind such as learning and problem-solving. Many AI models were proposed to be used side by side for this research so that the results are obtained faster. The main advantage of this approach was that the results were obtained faster as the models were utilized in parallel to one another but the disadvantage was the amount of time taken due to the complex nature of these algorithms.

By studying the above 5 research papers on breast cancer, understood that our algorithm can provide us with better accuracy to predict the cancer as well as can do it faster than them as the algorithm used is not that complex to deal with.

III. Methodology

By the help of latest technologies, the probability of breast cancer can be predicted in a human much efficiently and accurately. Machine learning will be used to predict breast cancer. The dataset of around 570 data entries (rows) and 32 attributes (columns) is obtained. By using multivariate regression, our machine will be trained to predict if the cancer is malignant or benign. A logistic regression algorithm will also be used to converge the attributes so that our machine can predict faster. 550 of our data entries will be used to train our machine and use the rest to test it so can be sure that our machine predicts the type of cancer correctly. All of our algorithms will be implemented in the python programming language by using machine learning libraries such as scikit-learn, TensorFlow and, so on.

The following steps will be used to get our machine to predict if the cancer is malignant or benign:

- The dataset from the UCI Machine Learning Repository will be used and do some cleaning of data (i.e. making the data suitable for our use by removing the empty columns and assigning numbers to text data like malignant is assigned to 1 and benign to 0) for prediction purpose.
- The number of malignant and benign cells in our dataset can be found and create a pair plot with respect to each feature so that can understand what feature has more effect in the prediction. To visualize the correlations better a heat map is used.
- After understanding the correlations, our predictor variable is decided and the target variable. And then split the data into training data (which in our case will be 75%) and testing data (25%). Then at the end scale the data.
- Then a function that contains 2 training models namely logistic regression model and decision tree model is made. The accuracies of these two models are then tested and proceed with the one which has maximum accuracy to predict the cancer.
- At last, the type of cancer is predicted and compare it to the actual values of our testing dataset.

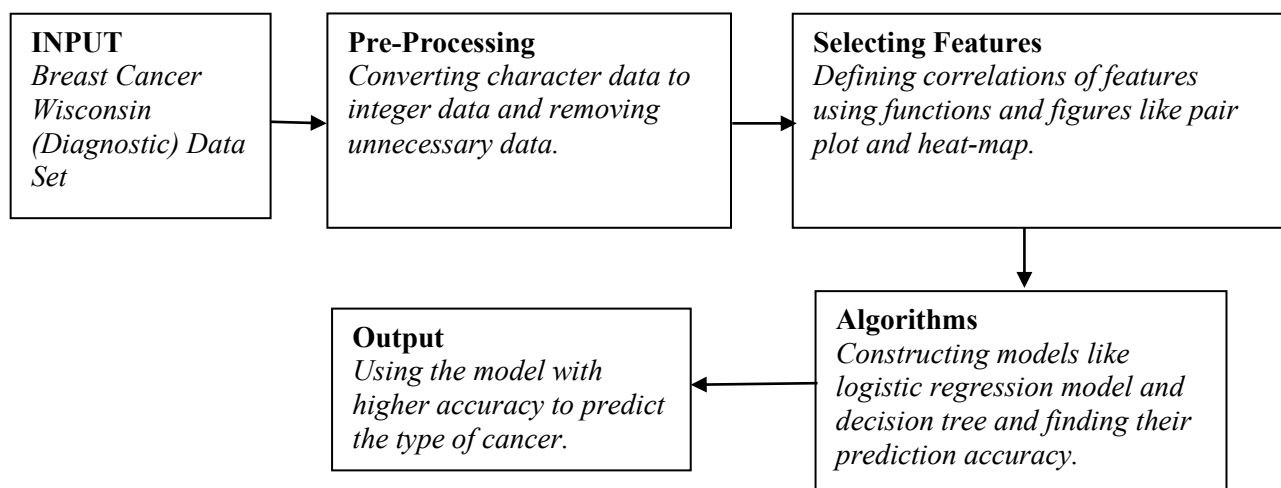


Fig. 3. Block Diagram for proposed model.

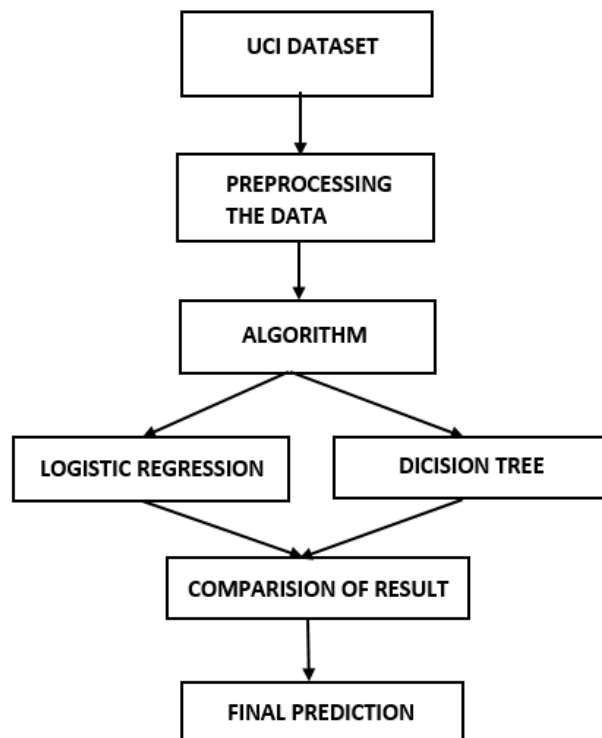


Fig. 4. Methodology Diagram

There are 2 algorithms used in our project:

- Logistic Regression
- Decision Tree

Logistic Regression:

Logistic Regression is an appropriate type of regression model when sure that our dependent variable is of binary type. Logistic regression is used to find the relationship between one dependent variable (which is binary in nature) and one or more independent variables.

Decision Tree:

Decision tree is a structure where each inner node represents a test on an attribute (such as if a coin when flipped will land on its tails or head), each of its branches represents the outcome of that test or attribute and each leaf node represents a class label (that is the decision taken after taking into account all the attributes).

Data Processing:

Converting the M (Malignant) and B (Benign) in the dataset to 1 and 0 resp. Scaling data is needed to normalize the range of each feature. Data splitting percentage is:

- Training Data: 75% (426 records)
- Testing Data: 25% (143 records)

Heat map:

The Heat Map helps us visualize the correlations better and gives the information about the influence of other features (columns) on the diagnosis column.

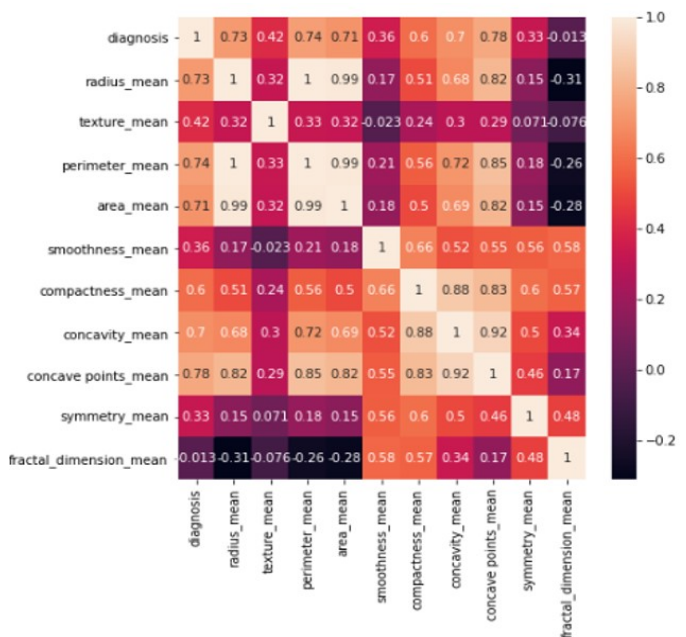


Fig. 5. Heat map

Affected and Normal Tumors comparison:

There are 569 records out of which 357 are noncancerous (Benign) and the remaining 212 are cancerous (Malignant). The following image shows the normal (Benign) and affected (Malignant) cell's comparison in our data set.

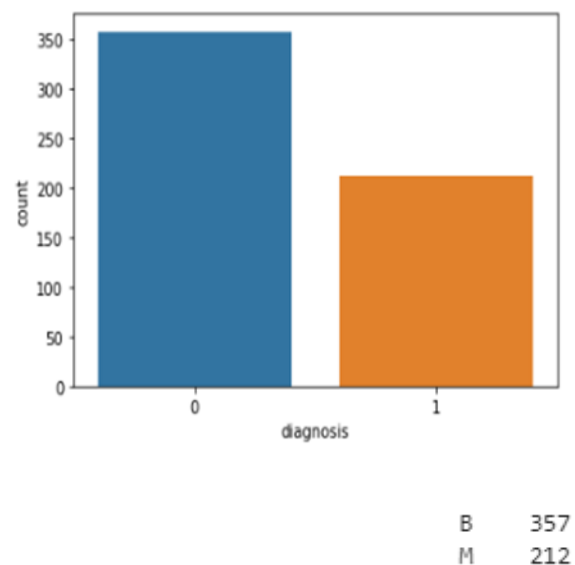


Fig. 6. Number of Benign and Malignant record

Pair Plot of diagnosis with other features:

The pair plot shows, visually what are the amount of malignant and benign cells respectively related to the features. This helps us see what features to take into consideration while predicting results. There are 569 records out of which 357 are noncancerous (Benign) and the remaining 212 are cancerous (Malignant). The following image shows the normal (Benign) and affected (Malignant) cell's comparison in our data set.

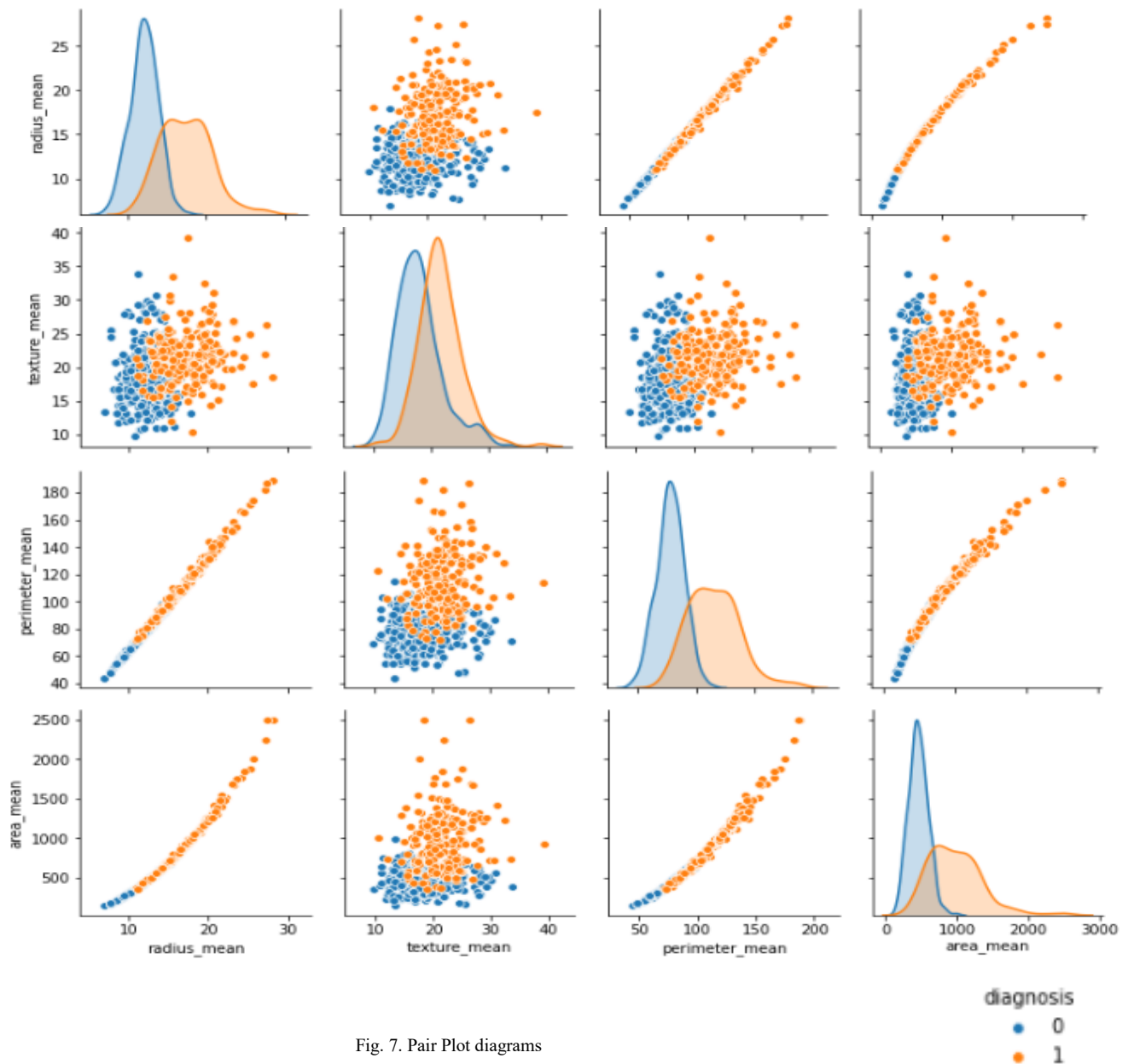


Fig. 7. Pair Plot diagrams

Accuracies:

These 2 algorithms (Decision Tree Classifier and Logistic Regression) are compared as both were predicted to generate predictions with high accuracy but needed that algorithm that has a higher accuracy among them so our predictions are more accurate, hence the decision tree classifier is chosen as it had slightly more accuracy than the logistic regression model. The following image shows the accuracy of the prediction of both the models and helps us decide which one to use.

```
[0]Logistic Regression Training Accuracy: 0.9906103286384976
[1]Decision Tree Classifier Training Accuracy: 1.0
[0]Logistic Regression Test Accuracy: 0.9440559440559441
[1]Decision Tree Classifier Test Accuracy: 0.951048951048951
```

Fig. 8. Accuracies of model

IV. Result

This shows the actual test value and the value predicted by our model side by side so that can get to see if the model is working as expected.

	Actual	Predicted
0	1	1
1	0	0
2	0	0
3	0	0
4	0	0
..	...	...
138	1	1
139	0	0
140	0	0
141	0	0
142	1	1
[143 rows x 2 columns]		

Fig. 9. Test values and Predicted values

V. CONCLUSION

In this research, two machine learning algorithms namely Decision Tree Classifier and Logistic Regression is implemented for prediction of breast cancer, and compared the accuracies of both to find which one of the two will be best suited for the prediction. Decision Tree Classifier is the best suited- algorithm for prediction as by using it had a pinpoint prediction accuracy [on “Breast Cancer Wisconsin (Diagnostic) Data Set”]. Hence, given the features according to that of this dataset, breast cancer can be predicted with almost pinpoint accuracy using our Decision Tree Classifier algorithm.

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